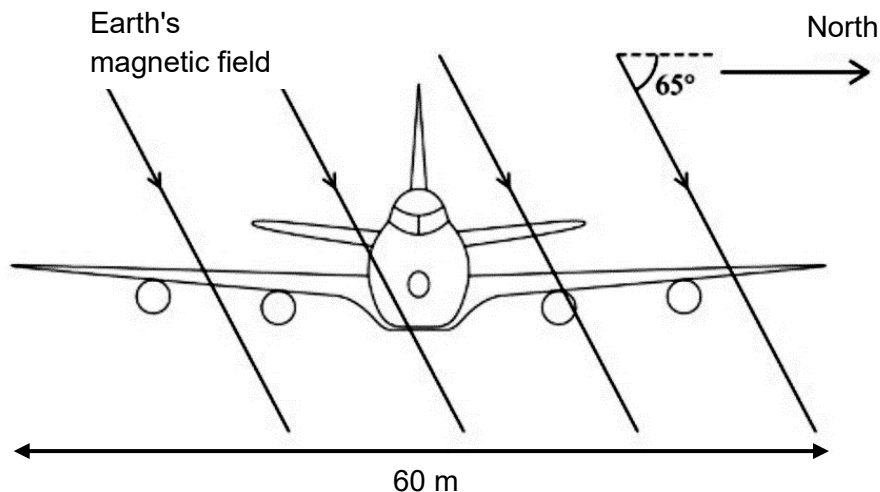


26. An aircraft, of wing span 60 m, flies horizontally at a speed of 150 m s^{-1} . The Earth's magnetic field in the region of the plane is $1.0 \times 10^{-5} \text{ T}$ and at an angle of 65° to the horizontal as shown below.



What emf is induced across the wing tips of the plane?

- A** 0.038 V **B** 0.082 V **C** 0.090 V **D** 9.0 V

Answer: B

Magnetic field perpendicular to wings = $1.0 \times 10^{-5} \sin 65^\circ$

Induced e.m.f. = $BLv = (1.0 \times 10^{-5} \sin 65^\circ) \times 60 \times 150 = 0.082 \text{ V}$